Identity Bridge Lite Paper

User-Sovereign, Privacy-Preserving Identity for Web3 and Web2

Michael Kerr, Jason Child
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"Users must control their identities. Privacy or celebrity as they prefer."
Christopher Allen, 10 Principles of Self-Sovereign Identity

Abstract

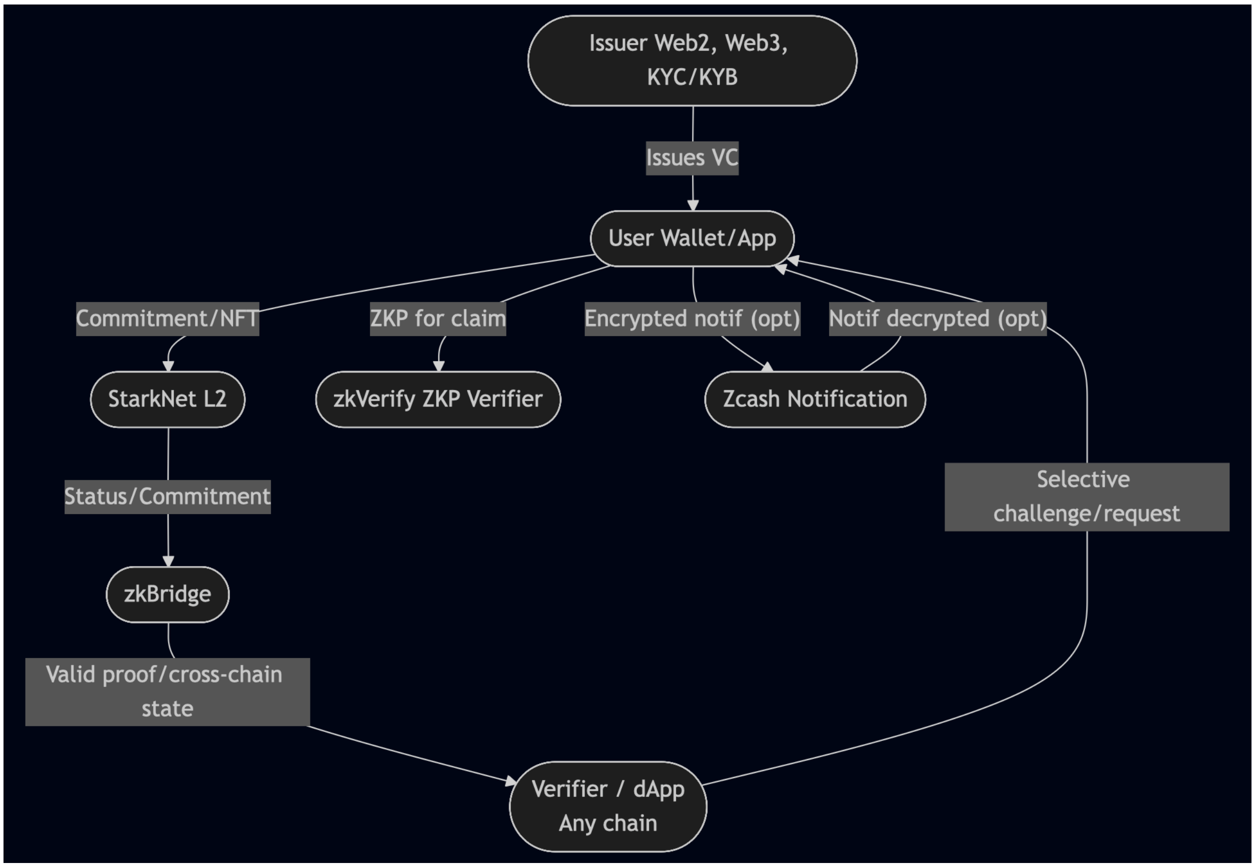
Identity Bridge is a full-stack, privacy-maximized digital identity protocol for the modern internet. The system leverages user-held verifiable credentials, powerful zero-knowledge proof systems (made practical and scalable by zkVerify), highly efficient and trustless cross-chain interoperability (via zkBridge), low-cost and programmable transactionality (using StarkNet L2), and privacy-enhanced notification (via Zcash shielded memos). The result is an ecosystem where users stay in control, privacy is rigorously enforced, and credentials flow safely and verifiably anywhere trust is required—without exposure or intermediaries.

Introduction

The goal of Identity Bridge is to place trust, privacy, and interoperability back into the hands of users. It links Web2 and Web3 identity in a cryptographically trustworthy, user-centric way, enforcing selective disclosure and privacy by design.

Protocol Components & Their Roles

Example Architecture



1. StarkNet L2

- **On-Chain Operations Hub**
  StarkNet L2 is the main platform for all on-chain logic, credential commitment anchoring, revocation, selective disclosure proofs, and programmable logic (NFTs, social recovery, credential registries). Its high throughput and near instant validity/finality offer near-free fees for users while inheriting the security of Ethereum L1.
- **Features**
  - Stores only minimal commitments—no raw data.
  - Secures credential state (mint, revoke, rotate).
  - Hosts NFT/non-transferable tokens to represent attested claims.

- Implements account abstraction for next-level UX (#multifactor, social recovery).

2. zkVerify

- **Zero-Knowledge Proof Verification Engine**
  zkVerify is used by smart contracts and off-chain infrastructures to validate advanced ZKPs efficiently and at minimal cost. It supports various proof systems, batch verification, and evolving cryptographic primitives.
- **Features**
  - Runs within smart contracts for low-cost, on-chain credential/attribute proofs.
  - Supports new circuits/types as identity requirements and privacy schemes evolve.
  - Works for both local user ZKPs and those relayed via zkBridge.

3. zkBridge

- **Trustless Cross-Chain State Relayer**
  zkBridge links StarkNet L2 (and other supported chains) with the rest of the multi-chain/web3 world. It cryptographically relays block headers, credential status updates, and ZKPs to destination chains using succinct zero-knowledge proofs—without oracles, multi-sig committees, or risks of centralized relayer attacks.
- **Features**
  - Any dApp or smart contract on a destination chain can trust credential status/attestation proven on StarkNet, with only a ZK proof and header—no trust in relayers needed.
  - Enables arbitrary cross-ecosystem use cases including identity-based airdrops, reputation, or compliance.
  - Preserves privacy, since only proofs/commitments are relayed—never credentials or underlying data.

4. Zcash (Privacy-Enhanced Notification Layer)

- **Supplementary Encrypted Notification**
  Zcash is used as an *opt-in supplementary privacy anchor* for notifications that a credential exists, is revoked, or otherwise updated—without revealing identity

links to public chains. The Zcash shielded memo field enables secure, off-chain notification between issuers and users, or privacy-centric relay between parties.

- **Features**
    - Encrypted, asynchronous, and unlinkable: Only the recipient can decrypt.
    - Ideal for “new credential received”, “revocation triggered”, or “status update” messages where privacy is at a premium.
    - Does *not* store credentials, serve as main registry, or act as a cross-chain bus.

End-to-End User Flow

1. Credential Issuance (Web2 → Web3)

- User attests to a Web2 provider (ex: Google, Socure). They authenticate, then generate a W3C Verifiable Credential, delivered directly/signature-backed to their device.
- Their wallet/app generates a cryptographic commitment and posts it as a “soulbound” NFT onto the StarkNet L2.

2. Proof and Verification

- When access to a service is needed, the user generates a ZKP locally (or in a TEE) about the credential or subset of claims.
- ZKProof is either verified on StarkNet via zkVerify for an on-chain dApp, or sent cross-chain with a succinct, relayed ZKP header via zkBridge.

3. Privacy and Selective Disclosure

- The verifier receives only the proof for the specific request—never underlying details (email, age, KYC doc, etc.), only that “the claim is valid and live”.

4. Notification (Zcash, Optional)

- Users/issuers can send status changes (e.g., “credential valid”, “revoked”, “updated”) encrypted via Zcash memos—ensuring privacy even for off-chain triggers.

5. Credential Lifecycle

- Users may revoke, rotate, or recover credentials using smart contract logic (guardians, social backup, or advanced threshold signatures) all on StarkNet—anchored, updateable, and always privacy-preserving.

Security, Incentives & Compliance

- **zkVerify and zkBridge** enable public, universal, cryptographic trust in every proof—no centralized attestors, no relayer games.
- **Token incentives** compensate infrastructure, paymaster gas, upgrades, and governance while enforcing solid economics for public goods and adoption.
- **All components** adhere to privacy law best practices: no public PII, auditability, and open participation for issuers and verifiers.

Prospective $SOVR Token

One Token, All Services

Identity Bridge will implement a unified payment model where users only need the *Sovereign* ($SOVR) tokens for all protocol operations. The system automatically handles backend conversions to STRK (for gas), ZEC (for privacy notifications), and payments to external services (zkVerify). This dramatically simplifies UX while creating strong $SOVR token demand.

Conclusion

Identity Bridge is infrastructure for the age of privacy. Every layer—zkVerify, zkBridge, StarkNet L2, user wallet, and privacy-enhanced Zcash notification—works together to anchor every proof in cryptography, never trust. User control is maximal, privacy is absolute, and access to proofs is available everywhere they're needed, never where they're not. This is the foundation for verifiable, user-owned digital identity for the next era of the internet.

Academic and Technical Citations

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